

Geometry

Guided Notes

Coordinate Geometry

Equation of a Circle

Guided Color-coded

Interactive Math Notebook Page

Circles in the Coordinate Plane

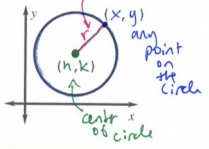
The Distance Formula is used to find an equation of a circle with center (h, k) and radius r .

- Let (x, y) be any point on the circle.
- Then the radius r is the distance from (h, k) to (x, y) .

Start w/ Distance Formula

$$r = \sqrt{(x-h)^2 + (y-k)^2}$$
$$r^2 = (x-h)^2 + (y-k)^2$$

Square both sides



STANDARD EQUATION OF A CIRCLE

In the coordinate plane, the standard equation of a circle with center at (h, k) and radius r is:

$$(x-h)^2 + (y-k)^2 = r^2$$

The x-coordinate of the center is h . The y-coordinate of the center is k .

Circles in the Coordinate Plane

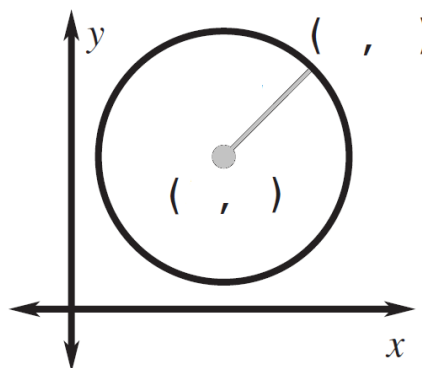
The **Distance Formula** is used to find an equation of a _____

with _____ (____, ____) and _____ .

- Let (____, ____) be any _____ on the _____.
- Then the _____ is the **distance** from (____, ____) to (____, ____).

$$\text{---} = \sqrt{(\text{---} - \text{---})^2 + (\text{---} - \text{---})^2}$$

$$\text{---} = (\text{---} - \text{---})^2 + (\text{---} - \text{---})^2$$



STANDARD EQUATION OF A CIRCLE

In the coordinate plane, the standard equation of a _____ with

_____ at (____, ____) and _____ is:

$$(\text{---} - \text{---})^2 + (\text{---} - \text{---})^2 = \text{---}^2$$

The ____-coordinate of the center is _____. The ____-coordinate of the center is _____.

Circles in the Coordinate Plane

The Distance Formula is used to find an equation of a circle

with center (h, k) and radius r .

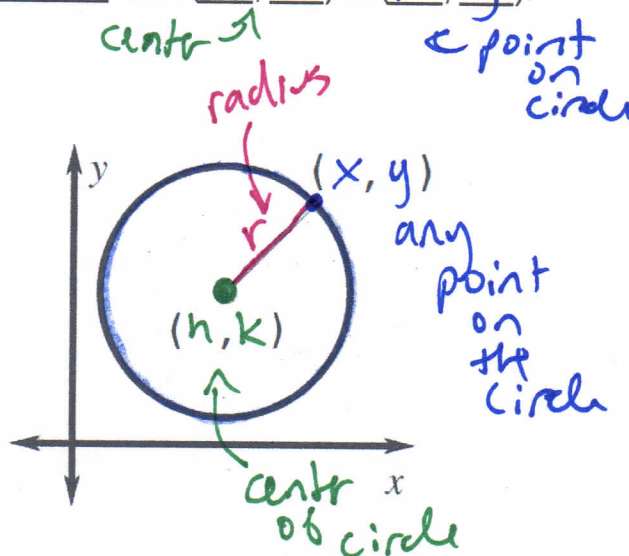
- Let (x, y) be any point on the circle.
- Then the radius r is the distance from (h, k) to (x, y) .

★ Start w/ Distance Formula
↓

$$r = \sqrt{(x-h)^2 + (y-k)^2}$$

$$r^2 = (x-h)^2 + (y-k)^2$$

★ Square both sides



STANDARD EQUATION OF A CIRCLE

In the coordinate plane, the standard equation of a circle with

center at (h, k) and radius r is:

$$(x-h)^2 + (y-k)^2 = r^2$$

Handwritten annotations: Blue arrows point from x and y to the text 'point on the circle'. Green arrows point from h and k to the text 'center of the circle'. A red arrow points from r to the text 'radius of the circle'.

The x-coordinate of the center is h . The y-coordinate of the center is k .



Thank you for downloading



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I hope this resource made your math lesson "Mathtastic".

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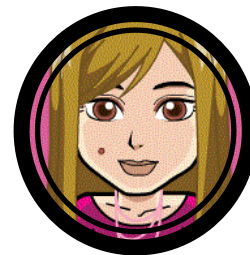
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